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Appropriateness and utility of a clinical decision support system at the digital front door



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Digital front doors combining symptom assessment and self-triage are increasingly used, yet real-world evidence on their clinical integration remains limited. This prospective post-market clinical follow-up evaluated a clinical decision support system embedded in routine care within a large private healthcare network. Adults completed a structured symptom assessment before consultation, and both participants and physicians completed predefined questionnaires. Treating physicians generally judged the urgency advice and assessment report to be appropriate. When the report was reviewed before the consultation, physicians reported greater preparedness and perceived efficiency gains. Participants generally reported complete symptom capture and improved preparedness. These findings provide prospective real-world evidence that a symptom assessment CDSS can generate appropriate outputs in routine care and may support both patient preparation and clinician workflow when integrated into practice.

Health systems face mounting pressure from population ageing, multimorbidity, and persistent workforce shortages. These trends increase demand while restricting the supply of clinical time, lengthening queues, and threatening timely access to appropriate care. Delayed or misdirected access has been associated with poorer outcomes, including avoidable hospitalisations and excess mortality, underscoring the need for scalable approaches that preserve both quality and safety^{1,2}.

Efforts to address these pressures can be grouped across three phases of the care pathway. Before the visit, patients require support to determine whether care is needed, where to seek it, and how urgently; in some cases, empowering them with reliable information may enable safe self-management. During the visit, clinicians benefit from tools that improve efficiency and quality of the encounter. High documentation burden remains a persistent challenge: in acute care, overcrowding, often driven by non-urgent presentations, forces clinicians to take and record histories under time pressure, with frequent interruptions that increase the risk of omissions³. Documentation quality varies with experience: junior physicians may take three times longer to complete histories, while senior physicians are faster but sometimes less complete⁴. Overstretching under these conditions is linked to higher error rates⁵. Structured, interactive, and pre-populated records have therefore been proposed as part of the solution, with

digital history-taking tools offering the potential to improve clarity, accuracy, and completeness while reducing burden⁶. After the visit, technologies for monitoring and asynchronous communication can reduce unnecessary attendances, support adherence, and promote continuity. The shared goal across phases is to optimise use of healthcare resources, enhance the experience of both patients and professionals, and improve outcomes.

Technology is increasingly being explored as a lever across these phases. AI-based scribe tools aim to reduce documentation burden and free time for patient interaction. Machine learning has been applied to clinical decision support and error reduction⁷. Large language models are being evaluated for summarisation, triage, and diagnostic support, though careful integration is required to ensure reliability⁸.

A particular class of tools with relevance at the entry point to care are the so-called digital front doors (DFD). These patient-facing, digitally enabled platforms combine symptom assessment, self-triage, booking, and sometimes remote consultations to guide people to the right service at the right time^{9,10}. While conceptually spanning all three phases, most implementations focus on the pre-visit stage and preparation for the consultation.

The evidence on DFDs based on clinical decision support system (CDSS) remains mixed. Systematic reviews report wide variation in diagnostic accuracy and appropriateness of urgency advice, with performance

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influenced by presentation type and evaluation method^{11,12}. Much of the literature is based on vignette studies or specialty-specific pilots, limiting generalisability. Real-world evaluations across broader patient populations and specialties remain scarce, although recent prospective studies have begun to explore patient-facing AI tools integrated into clinical workflows, primarily in controlled or single-site settings.

Against this backdrop, the ESSENCE (E-health Self Symptom assessment as a front door and facilitator of Care) post-market clinical follow-up study evaluates a regulated symptom assessment device integrated into routine practice in Portugal. In contrast to our companion analysis focusing on health-seeking behaviour¹³, this paper examines the appropriateness of the information provided and its impact on consultation quality and effectiveness, asking whether the tool supports safe and appropriate guidance, strengthens decision-making and care navigation, and enables more prepared and efficient consultations.

Results

Study population

A total of 1470 participants were enrolled, with a mean age of 38.5 years (SD 12.5, range 18–83), of whom 57.7% (848/1470) were female. Participants initially entered a median of two symptoms (IQR 1–3). Cases were grouped by clinical specialty according to their presentation. Participants completed a post-assessment survey, which provided information on the completeness of symptom entry, preparedness for consultation, and psychological effects of the assessment.

Physician surveys were completed for 163 consultations, in different settings: 13 in-person visits in the emergency department and 150 tele-consultations. Of the total, 103 surveys were completed after consultations in which the physician had not reviewed the report beforehand, and 60 were completed immediately after consultations in which the report was reviewed in advance. The latter group ($n = 60$) formed the basis for analyses of consultation efficiency and preparedness.

The participant inclusion process is outlined in a STROBE flow diagram (Fig. 1).

Reasonability of urgency advice

Physicians agreed that the urgency advice level associated with the most probable condition was reasonable in 74.1% of cases (120/161; 95% CI 67.4–80.1). No differences were observed by case specialty ($p = 0.535$), sex ($p = 0.237$) or age group ($p = 1.000$) (Table 1, Supplementary Table 1 and 1 C). For each case, physicians were asked whether all urgency advice levels provided by the assessment report, i.e. condition-specific urgency recommendations, were reasonable; in 77.6% (125/161; 95% CI 70.1–83.9) of cases the recommendations were considered reasonable. No significant differences were found with case specialty ($p = 0.107$), sex ($p = 0.710$), or age group ($p = 0.685$) (Table 1, Supplementary Tables 2 and 3).

The appropriateness of the care recommendations and comparison with actual participant behaviour were analysed by overall advice level, defined as the highest-acuity recommendation generated by the CDSS for each case (for example, if both self-care and emergency care were suggested, the case was categorised under emergency care). Table 2 summarises these results. Physicians rated the overall urgency advice as appropriate in 77.6% (95% CI 70.1–83.9), neutral in 13.7% (95% CI 8.5–20.2), and not appropriate in 8.7% (95% CI 5.0–13.9) (14/161). To understand whether advice judged not appropriate nonetheless resulted in safe user behaviour, the 14 cases in which the CDSS advice was rated not appropriate were examined and compared participants' actual care-seeking actions. In five of these cases (35.7%; 95% CI 12.8–64.9), participants' actions matched what physicians considered appropriate, indicating that some users sought care consistent with clinical expectations even when the CDSS advice was inaccurate (Table 2).

To further characterise the nature of the inappropriate advice, within the same 14 cases, the CDSS's highest (overall) advice level was compared with the physician's recommended care-seeking behaviour. Physician responses for what the appropriate behaviour in 3 levels ("Managing

symptoms at home or consulting a pharmacist", "Consulting a GP" or "Consulting a specialist" and "Emergency Room/Emergency") and were mapped onto the same eight-level CDSS advice scale (Supplementary Table 4). The CDSS recommendation coincided with the physician's behaviour recommendation in nine cases (64.3%; 95% CI 35.1–87.2) and was more cautious in five (35.7%; 95% CI 12.8–64.9). Importantly, no instances were identified in which the CDSS provided a less urgent recommendation than physicians considered appropriate, indicating that all discrepancies reflected conservative rather than unsafe guidance. It should be noted that physicians' recommended behaviours were captured using a simplified three-level categorisation (self-care, primary care, or emergency care), whereas the CDSS generated advice across an eight-level scale. Consequently, apparent agreements may partly reflect differences in scale granularity rather than true clinical agreement.

Appropriateness of the assessment report

Appropriateness of the assessment report was evaluated across multiple dimensions, including the reasonableness of suggested conditions, completeness of the symptom list, concordance of the main problem with the consultation focus, inclusion of the physician's most probable diagnosis, provision of additional useful information, and whether the report prompted further tests or flagged critical findings (Table 3).

In 73.0% of consultations (119/163; 95% CI 65.7–79.5), all suggested conditions were judged reasonable, while in a further 23.3% (38/163; 95% CI 17.2–30.5) at least some suggestions were considered reasonable; only 3.7% (6/163; 95% CI 1.4–7.9) were judged not reasonable. The symptom list was considered complete in 78.4% (127/162; 95% CI 71.2–84.4), and the main medical problem matched the consultation focus in 82.8% (135/163; 95% CI 76.1–88.2). Additional useful patient information was reported in 69.9% (107/153; 95% CI 62.1–77.0), the report prompted further diagnostic tests in 19.5% (31/159; 95% CI 13.8–26.5), and critical findings were flagged in 33.1% (52/157; 95% CI 26.0–41.0).

The physician's most probable diagnosis was included among the suggested conditions in 80.4% (131/163; 95% CI 73.5–86.0), with significant differences by specialty ($p = 0.035$), but no significant variation by sex ($p = 1.000$) or age group ($p = 0.191$) (Supplementary Table 5). Across most specialties, results were close to or above 80%, indicating generally strong correspondence between the CDSS suggestions and physicians' clinical reasoning.

Physicians were more likely to find the diagnosis included in the report when the consultation focused on the same symptoms or medical problem as those stated in the assessment: most probable diagnosis included in 87.4% (118/135; 95% CI 80.7–92.6) of such cases, compared with 46.4% (13/28; 95% CI 27.5–66.1) when the consultation addressed a different problem ($p < 0.00001$). This indicates that discrepancies in most likely diagnosis often arose when patients presented a different medical problem in the consultation than the one they had reported during the assessment.

Consultation efficiency and time

The mean consultation length, based on 59 valid responses, was 13.0 minutes (SD 5.4; median 14; range 4–28). Time savings during the consultation were reported in 66.1% (39/59; 95% CI 52.3–77.6), while 33.9% reported no time saved (20/59; 95% CI 22.4–47.7). Time was most often saved during the consultation itself (29/39; 74.4%; 95% CI 59.7–84.7). Time saving was significantly ($p = 0.033$) more often observed in clinical cases judged as easy (11/15; 73.3%; 95% CI 44.9–92.2) compared with moderate or difficult ones (10/26; 38.5%; 95% CI 20.2–59.4).

Physicians rated the usefulness of the condition suggestions for the consultation work favourably. Across 60 valid responses, 46.7% (28/60; 95% CI 33.7–60.0) judged the report to be very useful, 38.3% (23/60; 95% CI 26.1–51.6) as somewhat useful, and 15.0% (9/60; 95% CI 7.1–26.6) as not useful. Efficiency improvement was reported by 66.7% (40/60; 95% CI 53.3–78.3) (Table 3).

Efficiency improvements were more frequent when all suggested conditions were judged reasonable (72.3%, 34/47; 95% CI 57.4–84.4),

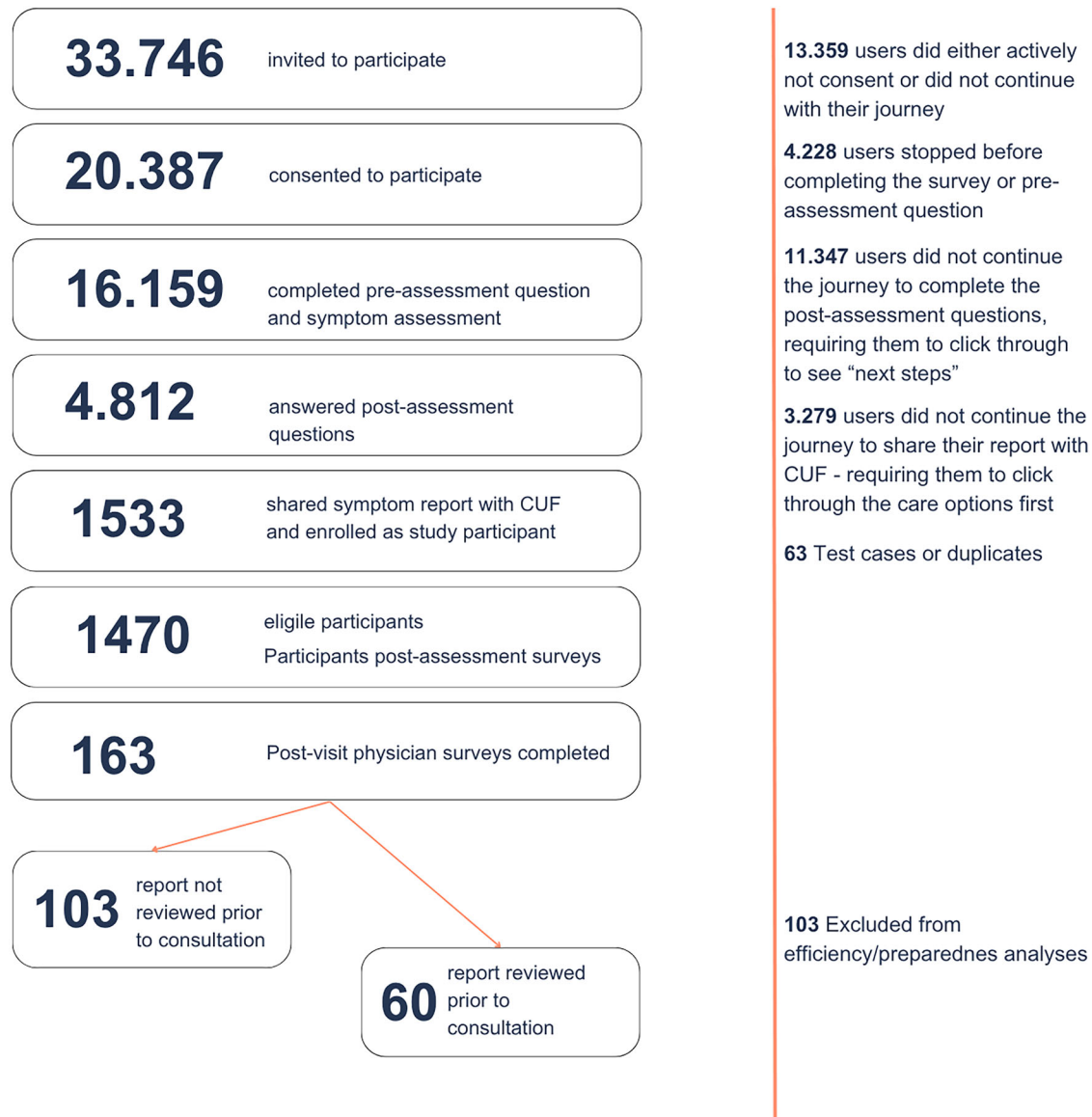


Fig. 1 | ESSENCE study workflow. This figure shows the sequence of steps in the study, including patient symptom assessment, generation of the CDSS report, physician consultation, and completion of participant and physician questionnaires.

compared with 60.0% (6/10; 95% CI 26.2–87.8) when only some were reasonable and 0.0% (0/3; 95% CI 0–70.8) when none were reasonable, although this trend did not reach statistical significance ($p = 0.066$). In contrast, perceived usefulness of the condition suggestions was a strong predictor of efficiency: improvements were reported in all cases rated very useful (100%, 28/28; 95% CI 87.9–100), compared with 52.2% (12/23; 95% CI 30.6–73.2) when somewhat useful and 0.0% (0/9; 95% CI 0–33.6) when not useful ($p < 0.0001$) (Supplementary Table 6).

Preparedness for the consultation was reported by 71.7% (43/60; 95% CI 58.6–82.5). This outcome was significantly associated with perceived usefulness of the suggested conditions ($p < 0.0002$) but not with their reasonableness ($p = 0.56$) (Supplementary Table 7).

Participant survey findings

In the post-assessment participant survey, across the three participant-reported endpoints (symptom capture in report, preparedness for consultation, and reduction of anxiety), a joint analysis was possible for those with non-missing answers, i.e. participants without skipped questions to all three items (1383/1470). In this cohort, 95.7% (95% CI 94.6–96.6) of participants reported at least one positive effect (22.2% [95% CI 19.9–24.7] in

one domain, 41.1% [95% CI 38.4–43.8] in two, and 32.3% [95% CI 29.8–35.0] in all three), while 4.3% (95% CI 3.4–5.4) reported no positive effect on any endpoint.

Overall, 80.5% (1183/1470; 95% CI 78.4–82.5) reported that all symptoms they wished to include were captured in the assessment, and 71.2% (1046/1470; 95% CI 68.7–73.6) felt prepared for their consultation (Supplementary Table 8). Preparedness was more common among those who reported all their symptoms (80.4% vs 65.7%; risk difference +14.7 pp, 95% CI 7.4–21.9; RR 1.22, 95% CI 1.10–1.36) (Supplementary Table 9).

One-third of participants (33.0%, 485/1470; 95% CI 30.5–35.6) reported feeling less anxious about their health after using the CDSS, while 30.9% (95% CI 28.4–33.4) reported not feeling less anxious and 29.0% (95% CI 26.6–31.5) reported no difference. A total of 7.1% did not respond to this item (skipped question). No significant differences in anxiety reduction were observed by sex or age group ($p = 0.49$ and $p = 0.46$, respectively).

Reporting all symptoms was associated with a higher likelihood of feeling less anxious after using the CDSS (37.8% vs 21.6%; risk difference +16.2 pp, 95% CI 9.7–22.7; RR 1.75, 95% CI 1.32–2.32). Similarly, those who felt prepared for the consultation after completing an assessment felt less anxious (43.9% vs 8.0%; risk difference

Table 1 | Appropriateness of urgency advice (* Agree = Strongly Agree or Agree)

		Most probable condition advice recommendation		All condition-specific recommendations	
		n/N (% agree)*	95% CI	n/N (% agree)*	95% CI
Speciality <i>p</i> = 0.535 (top), 0.107 (all)	Dermatology	7/10 (70.0)	34.8–93.3	5/10 (50.0)	23.7–76.3
	Gastroenterology	15/23 (65.2)	43.4–83.0	16/22 (72.7)	49.8–89.3
	Gynaecology	22/26 (84.6)	65.1–95.6	23/26 (88.5)	69.8–97.6
	Orthopaedics / Trauma	13/20 (65.0)	40.8–84.6	14/20 (70.0)	45.7–88.1
	Otorhinolaryngology / Ear Nose Throat	35/47 (74.5)	59.7–86.1	38/48 (79.2)	64.0–90.0
	Pulmonology	10/11 (90.9)	58.7–99.8	11/11 (100.0)	71.5–100
	Other	18/23 (78.3)	56.3–92.5	18/24 (75.0)	53.3–90.2
	Total	120/161 (74.1)	67.4–80.1	125/161 (77.6)	70.1–83.9
Sex <i>p</i> = 0.237 (top), 0.710 (all)	Female	76/97 (78.4)	68.8–86.1	76/96 (79.2)	69.6–86.9
	Male	44/64 (68.8)	56.1–79.9	49/65 (75.4)	63.3–85.2
Age <i>p</i> = 1.000 (top), 0.685 (all)	18–64	115/154 (74.7)	67.0–81.4	118/153 (77.1)	69.4–83.7
	≥65	5/7 (71.4)	29.0–96.3	7/8 (87.5)	47.3–99.7

Values are numbers and percentages of cases judged appropriate by physicians for both the top (most probable condition) and all urgency advice levels generated by the CDSS. Comparisons across specialties, sex, and age groups were performed using Pearson's χ^2 or Fisher's exact test as appropriate.

Table 2 | Appropriateness of care recommendation with actual participant behaviour comparison

Overall advice level	Appropriate n (%)	95% CI	Neutral n (%)	95% CI	Not appropriate n (%)	95% CI	In not-appropriate cases: Participant behaviour rated appropriate n (%) ^a
Self-care	3 (42.9)	9.9–81.6	4 (57.1)	18.4–90.1	0 (0.0)	0–35.4	–
Self-care + pharmacy support	19 (86.4)	65.1–97.1	3 (13.6)	2.9–34.9	0 (0.0)	0–14.8	–
Primary care in 2–3 weeks	3 (33.3)	7.5–70.1	5 (55.6)	21.2–86.3	1 (11.1)	0.3–48.2	Telemedicine (1): 0 (0.0)
Primary care in 2–3 days	33 (82.5)	67.2–92.7	2 (5.0)	0.6–16.9	5 (12.5)	4.2–26.8	Telemedicine (5): 2 (40.0)
Primary care same day	49 (81.7)	70.3–90.2	6 (10.0)	4.0–19.5	5 (8.3)	2.8–18.4	Telemedicine (5): 2 (40.0)
Primary care within 4 h	6 (100.0)	61.0–100	0 (0.0)	0–39.0	0 (0.0)	0–39.0	–
Emergency care	9 (81.8)	48.2–97.7	1 (9.1)	0.2–41.3	1 (9.1)	0.2–41.3	Telemedicine (1): 0 (0.0)
Call ambulance	3 (50.0)	11.8–88.2	1 (16.7)	0.4–64.1	2 (33.3)	4.3–77.7	Telemedicine (2): 1 (50.0)
Total (n = 161)	125 (77.6)	70.1–83.9	22 (13.7)	8.5–20.2	14 (8.7)	5.0–13.9	5 (35.7) (95% CI 12.8 – 64.9)

Cases were grouped by the highest-acuity recommendation (overall advice level) generated by the CDSS. Values show physician ratings of appropriateness (appropriate, neutral, not appropriate) and, for cases judged not appropriate, the patient's actual care-seeking behaviour and whether that behaviour was considered appropriate. Percentages are within each overall-advice category.

^aAlignment compares the action taken (Service Visited) with the physician's judgement of appropriate behaviour (Behaviour Appropriateness) for that case.

+35.9 pp, 95% CI 31.6–40.3; RR 5.51, 95% CI 3.70–8.21) (Supplementary Table 9).

Changes in intended care were related to changes in anxiety. Reduced anxiety was most common among those who de-escalated intended care (48.1%, 95% CI 43.2–53.0) and those who escalated care (44.0%, 95% CI 38.7–49.5), compared with 33.1% (95% CI 29.5–36.8) in the no-change group. Participants in the made-certain group reported reduced anxiety in 38.7% (95% CI 30.8–47.1) of cases, while those in the made-uncertain and stayed-uncertain groups reported lower rates (24.3% [95% CI 17.1–33.1] and 26.7% [95% CI 17.3–38.8], respectively) (Supplementary Table 10). Overall differences between groups were not statistically significant (*p* = 0.141).

Discussion

In routine clinical use within Portugal's largest private healthcare network, the CDSS generated urgency advice that treating physicians judged appropriate in approximately three quarters of cases, with no meaningful variation by specialty, sex, or age. Where advice was judged inappropriate, all discrepancies reflected conservative over-triage, the system erred toward

higher urgency rather than lower. No instances of under-triage were identified. This safety profile is consistent with a design principle that is appropriate for a pre-consultation tool: when uncertain, recommend more caution rather than less. Recent prospective evaluations of conversational diagnostic AI in primary care have similarly reported zero safety interventions across all patient interactions, reinforcing the view that appropriately supervised digital tools can operate within acceptable safety boundaries in real-world settings¹⁴.

Diagnostic concordance was high across most specialties with the physician's most probable diagnosis included among the CDSS suggestions in over 80% of cases. The notable exception was dermatology, where inclusion was considerably lower. This is consistent with the known limitations of text-based symptom assessment: dermatological diagnosis depends heavily on visual pattern recognition, which a system relying on reported history cannot replicate. This condition, rather than representing a system failure, defines the scope within which text-based CDSS can be expected to perform well. Concordance was substantially higher when the consultation addressed the same problem the patient had reported during the assessment, indicating that most apparent mismatches arose from

Table 3 | Physician-reported survey outcomes on assessment report appropriateness and consultation impact

Item	Response	n/N (%)	95% CI (%)
Appropriateness of the Assessment Report			
Conditions reasonability	All reasonable	119/ 163 (73.0)	65.7–79.5
	Some reasonable	38/ 163 (23.3)	17.2–30.5
	None reasonable	6/163 (3.7)	1.4–7.9
Symptom list complete	Strongly agree	57/ 162 (35.2)	28.0–43.0
	Agree	70/ 162 (43.2)	35.4–51.3
	Neutral	18/ 162 (11.1)	6.7–17.0
	Disagree	16/162 (9.9)	5.8–15.5
	Strongly disagree	1/162 (0.6)	0.0–3.4
Main problem matched consultation	Yes	135/ 163 (82.8)	76.1–88.2
	No	28/ 163 (17.2)	11.8–23.9
Most probable diagnosis included	Included	131/ 163 (80.4)	73.5–86.0
	Not included	32/ 163 (19.6)	14.0–26.5
Additional useful information	Yes	107/ 153 (69.9)	62.1–77.0
	No	46/ 153 (30.1)	23.0–37.9
Additional tests prompted	Yes	31/ 159 (19.5)	13.8–26.5
	No	128/ 159 (80.5)	73.5–86.2
Critical findings identified	Yes	52/ 157 (33.1)	26.0–41.0
	No	105/ 157 (66.9)	59.0–74.0
Impact of the Assessment Report in the Consultation			
Usefulness of condition suggestions	Very useful	28/60 (46.7)	33.7–60.0
	Somewhat useful	23/60 (38.3)	26.1–51.6
	Not useful	9/60 (15.0)	7.1–26.6
Efficiency improvement	Strongly agree	14/60 (23.3)	13.4–36.0
	Agree	26/60 (43.3)	30.6–56.8
	Neutral	11/60 (18.3)	9.5–30.4
	Disagree	7/60 (11.7)	4.8–22.6
	Strongly disagree	2/60 (3.3)	0.6–11.3
Physician preparedness for consultation	Strongly agree	9/60 (15.0)	7.1–26.6
	Agree	34/60 (56.7)	43.2–69.4
	Neutral	9/60 (15.0)	7.1–26.6
	Disagree	4/60 (6.7)	1.8–16.2
	Strongly disagree	4/60 (6.7)	1.8–16.2
Confidence in diagnosis	Strongly agree	14/60 (23.3)	13.4–36.0
	Agree	23/60 (38.3)	26.1–51.6
	Neutral	12/60 (20.0)	10.7–32.6
	Disagree	8/60 (13.3)	6.0–24.6
	Strongly disagree	3/60 (5.0)	1.0–13.9
Patient more engaged in consultation	Strongly agree	10/58 (17.2)	8.5–29.4
	Agree	35/58 (60.3)	46.6–73.0

Table 3 (continued) | Physician-reported survey outcomes on assessment report appropriateness and consultation impact

Patient more informed in consultation	Neutral	7/58 (12.1)	5.0–23.3
	Disagree	6/58 (10.3)	3.9–21.2
	Strongly disagree	0/58 (0.0)	0–6.2
	Strongly agree	14/59 (23.7)	13.7–36.6
	Agree	30/59 (50.8)	37.6–64.0
	Neutral	10/59 (16.9)	8.5–29.0
	Disagree	5/59 (8.5)	2.8–18.7
	Strongly disagree	0/59 (0.0)	0–6.1

Values are numbers and percentages of physicians selecting each response option. Comparisons across subgroups (specialty, sex, age) used Pearson's χ^2 or Fisher's exact test where applicable.

patients presenting a different concern at the visit rather than from inaccurate suggestions by the CDSS.

Among the subset of consultations where physicians reviewed the report before the encounter (60 of 163), the majority reported improved preparedness and efficiency gains, with time savings occurring most often during the consultation itself. However, the factor most strongly associated with perceived benefit was not whether the suggested conditions were judged reasonable, it was whether physicians found them useful for their clinical work. Efficiency improvements were reported in all cases where condition suggestions were rated very useful, compared with roughly half when rated somewhat useful and none when rated not useful. The same pattern held for preparedness.

This distinction matters. It suggests that for a pre-consultation CDSS, generating clinically plausible outputs is necessary but not sufficient. What determines whether the tool improves consultations is whether the information is structured and presented in a way that clinicians can act on within their workflow. Similar findings have emerged from studies of pre-visit conversational AI, where management plans from AI matched physicians in overall quality and safety but were rated lower for practicality and cost-effectiveness, dimensions that depend on contextual knowledge the AI did not have¹⁴. Together, these observations point toward a practical implication: optimising CDSS for real-world impact may require focusing not only on algorithmic accuracy but also on the format, integration, and actionability of the outputs.

Evidence from computer-assisted history-taking and structured digital templates supports this interpretation. Pre-populated and structured records have been shown to improve completeness and data quality compared with unstructured note-taking^{6,15}, and structured digital tools have been found to improve completeness and confidence among junior clinicians in particular^{16,17}. This study however was not designed to determine whether the report provided clinicians with information they would not otherwise have obtained through routine history-taking, or whether it changed the structure or duration of the consultation. Answering those questions would require objective workflow measures such as consultation timing, interaction analysis, or controlled comparison with unassisted consultations.

Despite the positive findings among physicians who reviewed the report, only 60 of 163 surveyed consultations (37%) involved pre-consultation report review. This low uptake is a key finding from an implementation perspective. It means that the majority of consultations in this evaluation did not benefit from the CDSS report, regardless of its quality.

This pattern is not unique to the ESSENCE study. In a recent prospective trial of conversational AI in an ambulatory primary care setting, 27% of providers did not review the AI-generated transcript before the urgent care appointment, despite it being made available prior to the consultation¹⁴. Behavioral barriers to adopting digital health tools are well

documented and include perceived time cost, unfamiliarity, and friction in locating information within existing clinical systems^{18,19}.

In our setting, the CDSS report was integrated into the CUF electronic health record, but its placement within the clinical workflow was not optimised for pre-consultation review. The report was accessible but not surfaced at the point where physicians typically prepare for a consultation, meaning that even with EHR integration, the information competed with other documentation for clinician attention. This distinction is important: the barrier to uptake was not the absence of integration but its design. Improving report visibility, e.g. through prominent placement within the consultation preparation view, scheduling alerts that flag when a patient has completed a pre-assessment, or automatic inclusion in the consultation summary, may be more effective than integration alone. The efficiency gains observed among physicians who did review the report provide a concrete basis for motivating these workflow adjustments.

From the patient perspective, the large majority reported that all symptoms they wished to include were captured, and most felt more prepared for the consultation after completing the assessment. One third reported feeling less anxious. Participants who reported complete symptom capture were more likely to feel prepared, and those who felt prepared were substantially more likely to report reduced anxiety, suggesting an association from structured self-assessment through preparedness to perceived reassurance.

These findings should be interpreted with caution. All patient outcomes were self-reported and were captured using single items rather than validated psychological instruments. The anxiety item was framed around reduction (“Did the assessment results make you less anxious?”), which may introduce response framing bias. Structured, clinically validated pre-consultation assessment may differ from unguided searching by contextualising symptoms within plausible diagnostic hypotheses and providing clear next steps, although this study was not designed to test that mechanism directly.

The ESSENCE study contributes to a growing body of evidence on AI-supported digital front doors, but its specific contribution lies in examining what happens at the interface between algorithmic output and clinical workflow. Most evaluations of digital triage and symptom assessment tools focus on output accuracy, whether the system generates correct diagnoses or appropriate urgency recommendations^{11,12}. Fewer studies track whether clinicians use the output or whether doing so changes the consultation. Our companion analysis, focusing on health-seeking behaviour of the ESSENCE study participants, showed that the same CDSS integration was associated with measurable shifts in patient care-seeking behaviour, with 59% of participants changing their care pathway and appropriate care increasing from 30% to 64%¹³. In that analysis, the strongest predictor of appropriate real-world behaviour was having an appropriate intention after the assessment, an intention that was recorded before any service options were shown, meaning it was shaped by the structured information in the assessment itself.

Taken together, these two analyses suggest that the CDSS adds value at two levels: by shaping patient decisions about where and when to seek care, and by structuring information that, when reviewed, supports clinician preparedness and efficiency. The binding constraint on the second pathway, as this study makes clear, is not algorithmic performance but workflow integration and clinician uptake. This also highlights the importance of ongoing real-world performance monitoring for AI-enabled digital tools, which may evolve over time through updates and therefore require continuous verification in practice²⁰.

Several limitations should be considered for careful interpretation of study results. Participants were adults using the myCUF application within a private healthcare network, likely favouring individuals with higher digital access and health literacy. The ≥65 age group comprised few participants, and certain specialties were represented by small numbers. These factors may limit the extent to which the findings reflect all patient populations. The study was conducted within a single, well-resourced private system with rapid access to primary and specialist care and digitally integrated booking

pathways. Whether the same patterns of clinician benefit and patient experience would emerge in public or resource-constrained settings, where access barriers, waiting times, and cost structures differ substantially, remains uncertain and would require dedicated evaluation.

A key design limitation is that only 60 of 163 physician surveys were based on consultations where the report was reviewed before the encounter. Workflow-related endpoints, including consultation efficiency, preparedness, perceived usefulness, and time savings, were captured only in these 60 consultations and should not be generalised to all consultations. The remaining 103 surveys used a shorter questionnaire focused on endpoints that can reasonably be assessed without prior report review, such as urgency advice reasonableness, completeness of the symptom list, and concordance between the report and the consultation. Several outcomes including perceived usefulness, preparedness, consultation efficiency, and time saved were self-reported and should be interpreted as perceived impact rather than objectively measured performance. These measures may be influenced by reporting bias and variability in interpretation across clinicians and participants. The study did not include objective timing data, interaction analysis, or controlled comparison with unassisted consultations, so causal claims about workflow improvement cannot be made.

The anxiety outcome was captured using a single self-report item framed around reduction. This limits interpretability: participants responding “No” may include those whose anxiety was unchanged as well as those whose anxiety increased, and the item does not allow these groups to be distinguished. Future evaluations should use validated instruments capable of capturing bidirectional change.

The study lacked a control group. Without a comparator, we cannot determine how much of the observed change in patient preparedness or anxiety would have occurred independently of the CDSS, for instance, through the natural course of preparing for a scheduled consultation. Similarly, the absence of a comparator limits the ability to attribute efficiency gains specifically to the report rather than to other aspects of the consultation context.

This prospective evaluation nonetheless has several strengths. It provides real-world evidence from routine integration of a CE-marked CDSS within the care pathways of a major healthcare network, combining both patient and clinician perspectives across multiple specialties. The predefined primary endpoints provide structured and replicable measures of CDSS performance. The distinction between consultations with and without pre-consultation report review, with transparent assignment of endpoints to each, addresses a common limitation in post-market evaluations where workflow feasibility constrains data collection.

The prospective PMCF design, aligned with ISO 14155:2020, supports methodological rigour. Finally, the explicit integration with a companion behavioural analysis¹³ allows the appropriateness findings reported here to be interpreted alongside evidence of real-world impact on care-seeking decisions, providing a more complete picture of how CDSS outputs translate into clinical practice.

The ESSENCE study demonstrates that structured, AI-supported symptom assessment can produce clinically appropriate outputs in routine practice, but also that generating good information is not the same as integrating it into care. Only a minority of physicians reviewed the report before the consultation, and the strongest driver of perceived benefit was not the accuracy of the suggestions but their usefulness in practice.

For patients, the assessment appeared to serve a preparatory and, for some, a reassuring function, though the limits of single-item self-report measures mean these effects require confirmation with more rigorous instruments. Whether these patterns hold in public systems, older populations, or settings with greater access barriers remains an open question. Answering it will require replication beyond well-resourced private networks and integration of objective workflow and outcome measures.

In summary, this study shows that a clinically validated CDSS can generate appropriate urgency advice and diagnostically relevant suggestions in routine care, and that when reviewed, its outputs are associated with perceived improvements in physician preparedness and consultation

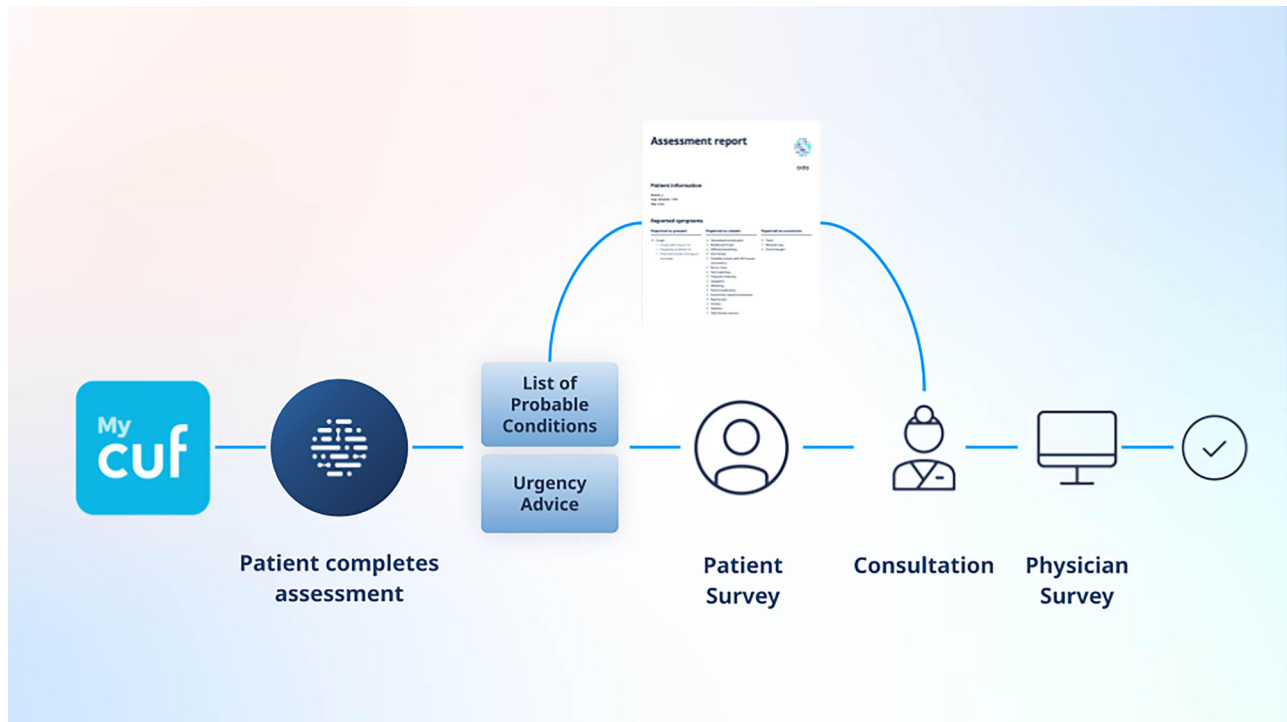


Fig. 2 | STROBE flow diagram of participant inclusion. This figure illustrates the inclusion process of participants in the ESSENCE study, including enrolment, exclusions, and final sample used for analysis. Figure created by the authors; icons sourced from Shutterstock under license.

efficiency. The findings further suggest that the impact of such systems depends less on diagnostic concordance alone and more on whether outputs are useful and integrated into clinical workflows. These results position workflow integration and clinician uptake, rather than algorithmic performance alone, as key determinants of real-world impact and provide direction for future optimisation and evaluation.

Methods

Study design and setting

This paper reports results from ESSENCE, a prospective quality-improvement evaluation of a digital clinical decision support system developed by Ada Health (Berlin, Germany) and implemented in Portugal's largest private healthcare network (CUF), integrated through the myCUF application. 'Ada Assess' (commonly known as 'Ada'), manufactured by Ada Health GmbH (Berlin, Germany), is a CE-marked Class IIa medical device under the EU MDR 2017/745.

Adults (≥ 18 years) using the myCUF app to complete a full assessment for themselves were eligible. Inclusion required completion of pre- and post-assessment questions, informed consent, and report sharing with a healthcare provider. Participants were instructed not to use the CDSS in emergency situations as per device intended purpose as described in the instructions for use and disclaimers. Enrolment ran from November 2023 to October 2024.

Data Collection Procedures

The full ESSENCE protocol has been reported previously¹³. The overall study workflow, including patient interaction with the system and subsequent physician use, is shown in Fig. 2.

Participants entered basic health information (e.g. age, sex) and described their presenting complaint. The CDSS then guided users through a series of dynamically selected questions, with each question determined in real time based on previously entered information. Based on the collected information, the system generated a list of probable conditions. These suggestions were derived from a probabilistic reasoning framework informed by a curated medical knowledge base, which encodes relationships

between conditions, symptoms, and risk factors. The model estimates the likelihood of potential conditions given the reported inputs and updates these estimates iteratively as new information is provided.

For each suggested condition, the CDSS assigned a condition-specific urgency level using a predefined eight-level scale (Self-care; Self-care with pharmacy support; Primary care in 2–3 weeks; Primary care in 2–3 days; Primary care on the same day; Primary care within 4 hours; Emergency care; Call ambulance). In addition, the system generated an overall urgency recommendation corresponding to the highest acuity level across all suggested conditions. This overall advice represented the guidance presented to the patient as the recommended next step in care and was linked to available care navigation options within the CUF system (e.g. video consultation, emergency care, ambulance). Immediately after completing the assessment, participants completed a survey capturing perceived completeness of symptom entry, preparedness for the consultation, and perceived psychological effects (including change in anxiety).

At the end of the assessment, the CDSS produced a structured report (Ada report example in Supplementary Information), which included a summary of reported symptoms and relevant medical history, a list of suggested conditions, and associated urgency advice levels (both condition-specific and overall recommendation). This report was made available to physicians within the electronic health record.

Physicians were instructed to review the Ada report prior to the consultation, conduct the consultation, and complete a post-consultation questionnaire. However, due to workflow constraints, pre-consultation review was not always feasible. For consultations where the report was reviewed before the encounter, physicians completed the full questionnaire, including items related to perceived usefulness, preparedness, efficiency, and time savings due to the report review. For consultations without prior report review, physicians completed a shorter post-consultation questionnaire limited to items that could reasonably be assessed without exposure to the report during the consultation, including appropriateness of urgency advice, reasonableness of suggested conditions, completeness of the symptom list, and concordance with the consultation focus and most probable diagnosis. Items requiring exposure to the report before

consultation (e.g. perceived impact on workflow or consultation dynamics) were analysed only in the subset of consultations where physicians confirmed reviewing the report in advance. The full physician post-consultation survey is provided in Supplementary Information, together with the rationale used to classify items based on whether they could be assessed without prior report review or required contemporaneous exposure to the report.

Participant and physician surveys combined Likert-type scales, categorical response options, and free-text fields.

Measurement framework and operational definitions

The ESSENCE study had two predefined primary endpoints: (i) appropriateness of the urgency advice and (ii) appropriateness of the assessment report. Endpoints were operationalised using structured participant and physician questionnaires, with survey and response options provided in Supplementary Information.

Appropriateness of urgency advice was assessed through treating physician ratings of whether the urgency advice levels presented in the Ada report (eight point scale described above) were reasonable based on the patient's symptoms, evaluated separately for (i) the most likely (top ranked) suggested condition and (ii) the advice levels across all suggested conditions. These assessments reflect the treating physician's clinical judgement.

Appropriateness of the assessment report was evaluated across pre-specified dimensions captured by physician survey items, including reasonableness of suggested conditions given the patient's presentation, completeness of the symptom list, concordance between the report and the consultation presentation, and inclusion of the physician's most probable diagnosis among the suggested conditions. For the diagnosis inclusion item, the questionnaire defined inclusion to encompass an exact match, an alternative name for the same condition, a more precise form of the suggested condition, a condition within the same umbrella term, or a related subtype.

For the physician survey, the terms *reasonableness* and *appropriateness* refer to structured clinical judgements made by the treating physician, anchored to the survey prompts and the report content. In this manuscript, *preparedness* reflects self reported perceptions of readiness for the consultation as recorded in the participant and physician questionnaires, and *efficiency*, *usefulness*, and *time saved* reflect self reported perceptions of impact on the consultation process. Each consultation was rated by a single treating physician; inter-rater agreement was therefore not assessed because there were no independent duplicate ratings per consultation.

Perceived anxiety change was included as a participant reported outcome to capture potential psychological impact of using the symptom assessment as a digital front door, including potential unintended effects, within the PMCF evaluation of device safety and benefit in routine use. Perceived anxiety change was assessed using the participant item "Did the assessment results make you less anxious?" with response options "Yes," "No," "No difference," and "Skip" (Supplementary Information).

Data analysis

Sample size considerations for the primary endpoint analysis were pre-specified in the ESSENCE protocol (ClinicalTrials.gov: NCT06846957 registry date: February, 26th, 2025). Recruitment for the primary endpoint analysis was planned to continue until the target number of participants with a follow up consultation with a participating physician was reached. The protocol includes the underlying hypothesis framework for advice appropriateness and the planned sample size calculation.

Survey responses were analysed descriptively using pre-defined response categories. Five-point Likert items were summarised using the original response distribution and, where required for endpoint analyses, dichotomised as positive (Agree or Strongly agree) versus non-positive (Disagree, or Strongly disagree). Participant items with an explicit Skip option were treated as missing for that item. For analyses comparing paired participant-reported outcomes, pre-specified positive and negative response definitions were applied and only participants with non-missing responses to both relevant items were included; denominators therefore vary across

analyses according to item-level missingness. No imputation of missing data was performed.

Continuous variables were summarised as means with standard deviations or medians with interquartile ranges, as appropriate, and categorical variables were summarised using frequencies and percentages. Proportions are presented with 95% confidence intervals calculated using the Wilson score method. Risk differences (percentage points) were calculated with 95% confidence intervals using the Wald method. Risk ratios were calculated on the log scale with 95% confidence intervals obtained by back-transformation; where any cell count was zero, a Haldane–Anscombe correction was applied.

Comparisons across subgroups (specialty, sex, and age group) were performed using Pearson's χ^2 test or Fisher's exact test, as appropriate; Fisher's exact test was used when expected cell counts were <5. All hypothesis tests were two-sided with a significance level of 0.05. For specialty-level analyses, specialties with fewer than ten consultations were grouped under Other to enable meaningful comparison. All analyses were conducted in Python (v3.10).

Data management and ethics

Data was securely stored in a validated electronic system (Teamscope) using unique identifiers. Ethical approval was granted by the Comissão de Ética para a Investigação Clínica (No. 2204JJ351 and No. 2309JJ660). The study was registered at ClinicalTrials.gov (NCT06846957) - registry date: February, 26th, 2025, and complied with the Declaration of Helsinki and ISO 14155:2020 guidelines. STROBE guidelines were followed throughout.

Data availability

The original data generated and analyzed in this study are included in the article and Supplementary Material. Data collected for the study will be made available to others after publication upon reasonable request, subject to approval. Requests for access should be made to the study team at Ada Health (science@ada.com). After approval, a signed data sharing agreement will be required before data release.

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Author contributions

F.C., S.G., P.F. and M.S.M. were responsible for conceptualisation and methodology. P.F., M.P.P., F.D.L., F.C., N.K., T.M., A.P. and M.S. contributed to investigation, project administration, supervision, validation, software and data curation. A.P. led writing the original draft in collaboration with all co-authors. A.L. contributed to project administration, supervision and writing - review and editing. All authors had full access to aggregated data in the study and contributed to the decision to submit the manuscript for publication.

Competing interests

A.L. is employed by Ada Health. A.P., N.K., M.S. and T.M. are former employees of Ada Health. S.G. and F.C. are consultants for Ada Health, and S.G., T.M., A.P. and F.C. hold share options in the company. S.G. declares a nonfinancial interest as an Advisory Group member of the EY-coordinated “Study on Regulatory Governance and Innovation in the field of Medical Devices” conducted on behalf of the Directorate-General for Health and Food Safety (SANTE) of the European Commission. S.G. declares the following competing financial interests: he has or has had consulting relationships with Una Health GmbH, Lindus Health Ltd., Flo Ltd, ICURA ApS, Rock Health Inc., Thymia Ltd., FORUM Institut für Management GmbH, High-Tech Gründerfonds Management GmbH, Prova Health Ltd, Directorate-General for Research and Innovation Of the European Commission. T.M. declares the following competing financial interests: he has or has had consulting relationships with Suvera & iPlato Healthcare. F.C. declares the following competing financial interests: she has or has had consulting relationships with Flo Ltd and Amboss SE. The remaining authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41746-026-02711-5>.

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